Miles Jennings

562-213-3013 | mjennin1@uci.edu | linkedin.com/in/miles-jennings | milesj-portfolio.app

Education

University of California Irvine — Irvine, CA
B.S Computer Engineering — Deans Honor List — *Sept. 2022 – June 2026*
Relevant Coursework: Operating Systems, Digital Systems, OOP, Data Structures & Algorithms, Computer Organization, Computer Networks
Organizations: HyperXite, FUSION, FASAE

Experience

Test Engineer Intern — Buena Park, California
Leach International Corp. — *March. 2026 – Present*

- Troubleshot and repaired test equipment by analyzing existing system architectures and adapting to legacy setups, eliminating production bottlenecks and ensuring on-time delivery
- Developed a LabVIEW-based capacitor test system (LCR6002 + 5-channel mux) for parallel validation of 350 components, eliminating outsourced testing and reducing company costs
- Built and maintained hardware test setups using oscilloscopes, multimeters, and power supplies to perform functional and performance testing on electrical components

HyperXite – SpaceX Hyperloop — Irvine, California
Control Systems Engineer — *Aug. 2025 – Present*

- Designed, tested, and optimized the Hyperloop pod's control system hardware using an STM32H753ZI microcontroller programmed in **C/C++**, interfacing with thermistors, time-of-flight sensors, and INA219 current sensors while evaluating components for performance and cost improvements over prior pod iterations
- Built ESP32-based bidirectional communication system using ESP-NOW for real-time control and telemetry between pod and station
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI

P2S Technology Intern — Long Beach, California
P2S Engineering — *June 2025 – March 2026*

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Improved design accuracy and project timelines by reducing regulatory revision cycles by 11% through proactive 3D modeling and cross-functional coordination
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs to ensure reliable infrastructure design

Projects

AudioVisor | *C, PDM, nRF52840, TypeScript, Vite, Supabase* — Sept. 2025 – March 2026

- Built an audio-reactive visor using an nRF52840, utilizing **C** to receive and handle digital microphone input through **PDM** communication for real-time sound magnitude/direction detection
- Developed a real-time web dashboard for the AudioVisor system to visualize and analyze audio data, achieving **84%** accuracy in detecting sound direction and amplitude during initial trials
- Researched and evaluated components to satisfy sensitivity, latency, and power needs, guiding system architecture and part selection

Building Management System | *Python, Raspberry Pi, LCD, DHT-11* — April 2025 – June 2025

- Designed and constructed a fully functional Building Management System (BMS) on Raspberry Pi using real-time data from DHT-11 and PIR sensors to control HVAC, ambient lighting, fire alarms, and security systems
- Programmed an interactive LCD interface and logging system to display live status updates, trigger event-based alerts, and ensure time-stamped logs with synchronization and interrupt-driven input handling

Technical Skills

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS Assembly, MATLAB

Software/Tools: Revit, AutoCAD, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies